

Chapter 6 Fluid resistance and energy loss

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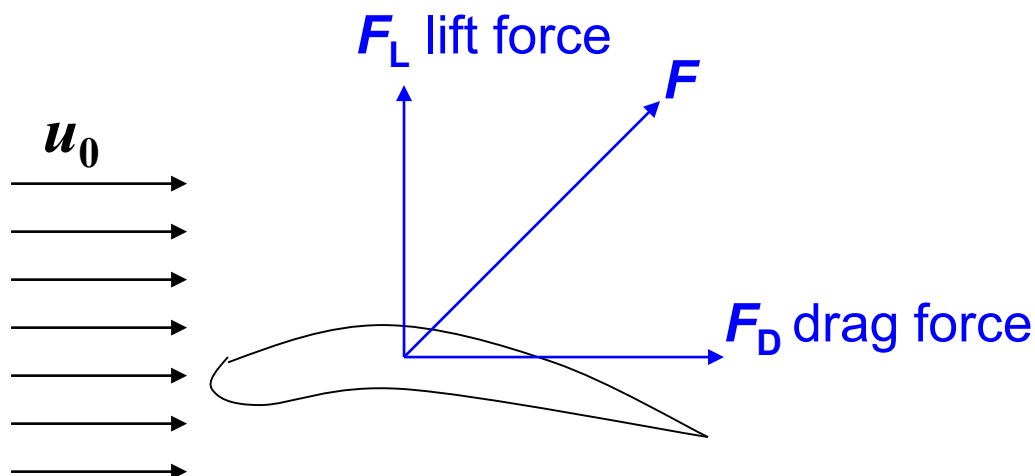
-  1. Characteristics of flow around an object
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Section 9 Drag due to flow around an object

1. Characteristics of flow around an object

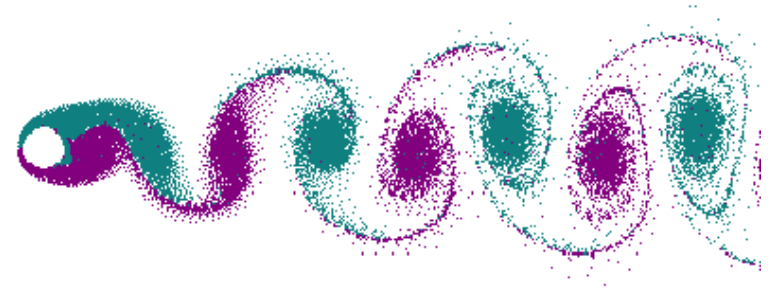
When a fluid and a solid submerged in it are in relative motion, the fluid forces acting on the solid contain:

- F_D parallel to the flow direction, called drag force (绕流阻力) ;
- F_L perpendicular to the flow direction, called lift force (升力) .

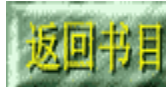


Section 9 Drag due to flow around an object

- The flow around a bluff object (钝体或阻流体) may induces fluid vibration.
- When Re reaches a certain value, the boundary layer separation occurs, creating alternating vortices on both sides of the downstream of the object. This is called the **Karman vortex street** (卡门涡街) .



- If the frequency of the vortex street is the same as or close to the natural frequency of the bluff object, resonance will occur and cause damages to structures.
- E.g., the vibration of risers (立管) and the breeze of sea-crossing bridge cables are all resulted from this phenomenon.



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Section 9 Drag due to flow around an object

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2. Calculation of drag force

$$F_D = C_D \rho \frac{u_0^2}{2} A$$

where,

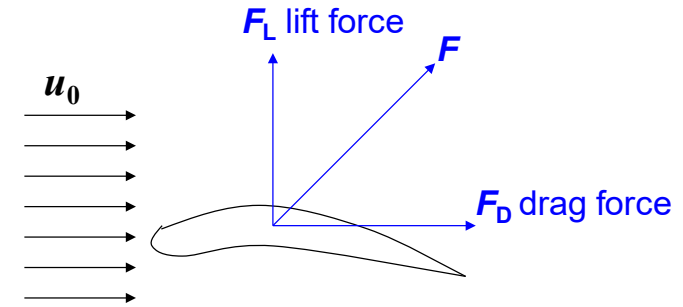
ρ : Fluid density;

u_0 : Relative velocity of undisturbed incoming fluid and the solid object;

A : Cross-sectional area of the solid object that faces the incoming fluid;

C_D : **Drag force coefficient**, value mainly depending on the shape of the object and Re . The C_D value of **small spheres** can be calculated as:

$$C_D = \begin{cases} \frac{24}{Re} & (Re < 1.0) \\ \frac{13}{\sqrt{Re}} & (1.0 < Re < 10^3) \\ 0.44 & (10^3 < Re < 2 \times 10^5) \end{cases}$$



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3. Sinking velocity of a sphere

Flow resistance when flow past a sphere

$$F_D = C_D \rho \frac{u_0^2}{2} A$$

$$C_D = \frac{24}{Re}$$

$$F_D = 3\pi\mu d u_0 \quad (Re < 1.0)$$

where, d ——sphere diameter; u_0 ——sinking velocity of sphere.

Analyzing the forces applied on the object, gives the limit velocity of the sinking of spherical object:

$$\sum F = 0 \quad F_{\text{浮}} + F_D = G$$

$$\frac{1}{6}\pi d^3 \cdot \rho g + 3\pi\mu d u_0 = \frac{1}{6}\pi d^3 \cdot \rho_s g$$

The **limit velocity** of the sinking of spherical object is:

$$u_0 = \frac{d^2}{18\mu} (\rho_s g - \rho g)$$

where ρ_s and ρ are the densities of the sphere and fluid, respectively.

The formulation can be used to design sedimentation tanks, dredging and sewage treatment analysis.



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1. Characteristics of two fluid regimes, i.e. laminar and turbulent (mixing, orderliness of particle movement, relationship between head loss and flow rate)

2. Identification criterion for laminar and turbulent flow: lower critical Reynolds number Re_c . For pipe flow, $Re_c < 2300$ is laminar flow. Re_c depends only on the boundary shape (i.e. the cross-sectional shape of fluid flow)

3. Basic equations for uniform flow : $\tau_0 = \rho g R J$ $\tau = \rho g R' J$

4. Incompressible steady uniform laminar flow

The velocity of circular pipe flow $u = \frac{\rho g J}{4\mu}(r_0^2 - r^2)$ follows the paraboloid of revolution.

$u_{\max} = \frac{\rho g J}{4\mu} r_0^2$ ($r = 0$) $v = \frac{1}{2} u_{\max}$ Maximum flow velocity is twice of the mean velocity.

$h_f = \frac{64}{Re} \frac{l}{d} \frac{v^2}{2g}$ Head loss is proportional to the first power of flow velocity,
Resistance coefficient $\lambda = 64/Re$.

5. Characteristics of turbulence: disorder, dissipating energy, diffusion.

Time average for turbulence quantities. Instantaneous velocity = time-averaged velocity + fluctuating velocity



Characteristics of fluid flow in circular pipes

Flow regime	Identification for flow regime	Identification for flow zone	Velocity distribution	Primary action forces	Factors affecting resistance	Head loss
Laminar flow	pipe flow: $Re < 2300$	No partition (不分区)	Parabolic distribution, non-uniform	Viscous force $\tau_0 = \mu \frac{du}{dy}$	$\lambda = f(Re)$ not related to $\frac{\Delta}{d}$	$h_f \propto v^1$
Turbulent flow	pipe flow: $Re > 2300$	Smooth turbulent flow	a. Linear distribution in viscous sublayer; b. Logarithmical or exponential distribution in other regions	Turbulent shear stress $\tau_0 = \tau_v + \tau_t$ $= (\mu + \eta) \frac{du}{dy}$ $\eta = \rho l^2 \frac{du}{dy}$	$\lambda = f(Re)$ not related to $\frac{\Delta}{d}$	$h_f \propto v^{1.75 \sim 2.0}$
		Transitional turbulent flow			$\lambda = f(Re, \frac{\Delta}{d})$	
		Rough turbulent flow			$\lambda = f(\frac{\Delta}{d})$ not related to Re	

6. Turbulent shear stress: $\tau = \tau_v + \tau_t = \mu \frac{d\bar{u}}{dy} + \rho l^2 \left(\frac{d\bar{u}}{dy} \right)^2 = (\mu + \eta) \frac{d\bar{u}}{dy}$

7. Velocity distribution for turbulent flow

a . Near the wall : $\frac{\bar{u}}{u_*} = \frac{u_* y}{\nu} \quad (y \leq \delta_1)$ linear distribution

b . Turbulence core region : $\frac{\bar{u}_x}{u_*} = \frac{1}{\kappa} \ln y + C \quad (y > \delta_1)$ logarithmic distribution

Thickness of viscous sublayer $\delta_l = \frac{32.8d}{Re\sqrt{\lambda}}$ decreases with the increase of Re

8. Energy loss $h_w = \sum_{i=1}^m h_{fi} + \sum_{k=1}^n h_{jk} \quad h_f = \lambda \frac{l}{d} \frac{v^2}{2g}, h_j = \zeta \frac{v^2}{2g}$

9. Solution of the frictional resistance coefficient λ

- Nikuradse's diagram. According to how λ relates to Δ and Re , the flow is divided into five zones (laminar flow, transition from laminar to turbulent flow, smooth turbulent flow, transition turbulent flow, rough turbulent flow).

- Colebrook–White equation
$$\frac{1}{\sqrt{\lambda}} = -2 \lg \left(\frac{2.51}{Re \sqrt{\lambda}} + \frac{\Delta}{3.7d} \right)$$

- Barr formula
$$\frac{1}{\sqrt{\lambda}} = -2 \lg \left(\frac{5.1286}{Re^{0.89}} + \frac{\Delta}{3.7d} \right)$$

- Chezy formula
$$v = C \sqrt{RJ} \quad C = \sqrt{\frac{8g}{\lambda}}$$

$$C = \frac{1}{n} R^{1/6} \quad \text{—— Manning formula}$$

10. Characteristics of boundary layer:

- a. Boundary layer thickness is a finite value;
- b. Boundary layer thickness increases along the flow direction;
- c. Boundary layer includes laminar flow boundary layer and turbulent boundary layer;

d. In the boundary layer:

Viscous sublayer – viscous shear stress dominants and turbulent shear stress is negligible.

Turbulent core region – both viscous and turbulent shear stresses be considered.

$$\tau = \mu \frac{du}{dy} + \rho l^2 \left(\frac{du}{dy} \right)^2$$

Outside the boundary layer: Velocity gradient is zero, no viscous force → flows can be solved by ideal/potential flow theories.